

BIOL 220 Probability

Learning Objectives

1. Determine if events are mutually exclusive
2. Determine if events are independent

Define probability

The probability of an event is the proportion of times that event would occur if we repeated a random trial over and over again under the same conditions. Probability always ranges between 0 and 1

Estimating probability

The probability of an event is the proportion of times that event would occur...

What's the probability I go to the beach and see a honu?

	December	January
Visits to beach	11	7
Seen a honu	4	2

Estimating probability

What's the probability I go to the beach and see a honu?

probability is (number of events) / (number of trials)

```
dec_prob <- 4 / 11
```

```
dec_prob
```

```
## [1] 0.3636364
```

```
jan_prob <- 2 / 7
```

```
jan_prob
```

```
## [1] 0.2857143
```

combine info using mean

```
mean(dec_prob, jan_prob)
```

```
## [1] 0.3636364
```

Rules of probability

1. Probability is always between 0 and 1
2. The sum of all the probabilities of all the possible outcomes always equals 1

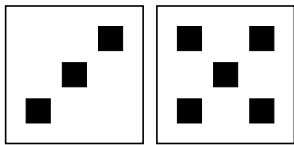
Rules of probability

1. Probability is always between 0 and 1
2. The sum of all the probabilities of all the possible outcomes always equals 1
3. We can *add* the probabilities of mutually exclusive outcomes

the orange and red regions are mutually exclusive

Rules of probability

1. Probability is always between 0 and 1
2. The sum of all the probabilities of all the possible outcomes always equals 1
3. We can *add* the probabilities of **mutually exclusive** events
4. We can *simply multiply* the probabilities of **independent** events



the event of the second die roll is **independent** of the first die roll, so the probability of a 3 for the first and a 5 for the second is

$$\frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Rules of probability

Adding and multiplying probabilities have special vocab:

Adding is an “OR” situation:

The mean is less than -2 OR greater than -1.7

$$Pr(\bar{y} < -2) + Pr(\bar{y} > -1.7) = 0.05$$

Rules of probability

Adding and multiplying probabilities have special vocab:

Multiplying is an “AND” situation:

The first die is 3 AND the second is 5

$$Pr(1^{st} \text{ die} = 3) \times Pr(2^{nd} \text{ die} = 5) = \frac{1}{36}$$

Rules of probability

Adding and multiplying probabilities have special vocab:

Adding is an “OR” situation:

The mean is less than -2 OR greater than -1.7

$$Pr(\bar{y} < -2) + Pr(\bar{y} > -1.7) = 0.05$$

The first die is 3 OR the second is 5

$$Pr(1^{st} \text{ die} = 3) + Pr(2^{nd} \text{ die} = 5) = \frac{1}{3}$$

Rules of probability

1. Probability is always between 0 and 1
2. The sum of all the probabilities of all the possible outcomes always equals 1
3. We can *add* the probabilities of **mutually exclusive** events
4. We can *simply multiply* the probabilities of **independent** events
5. We can *add* the probabilities of **independent** events

Rules of probability

1. Probability is always between 0 and 1
2. The sum of all the probabilities of all the possible outcomes always equals 1
3. We can *add* the probabilities of **mutually exclusive** events
4. We can *simply multiply* the probabilities of **independent** events
5. We can *add* the probabilities of **independent** events
6. What about multiplying **non-independent** probabilities???

Non-independent events

You go catch a fish and...

A

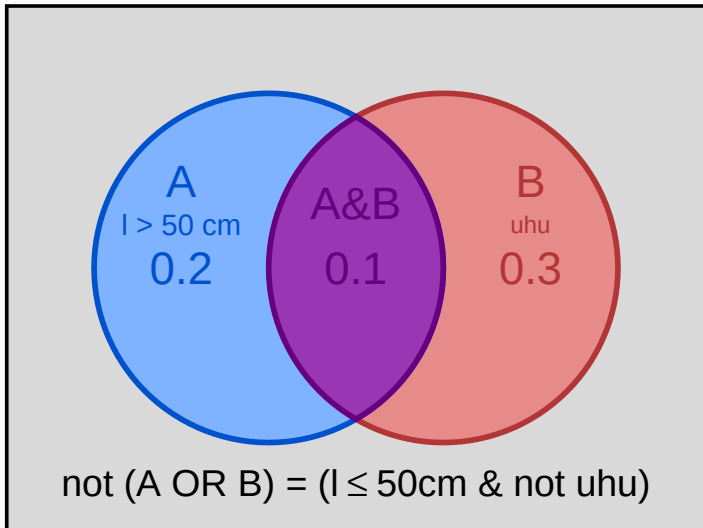
Length is greater than 50 cm

B

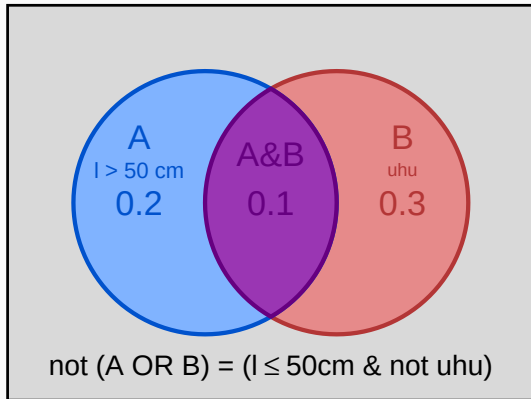
It's an uhu



Non-independent events



Non-independent events



Calculating probabilities

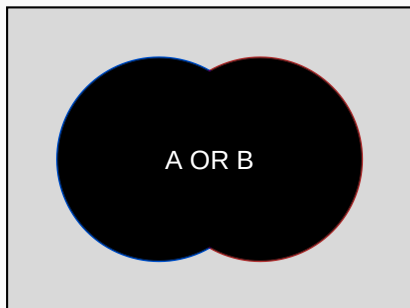
$$Pr(A) = 0.2$$

$$Pr(B) = 0.3$$

$$Pr(A \text{ \& } B) = 0.1$$

$$Pr(A \text{ OR } B) = ??$$

Non-independent events



Calculating probabilities

$$Pr(A \text{ OR } B) \neq Pr(A) + Pr(B)$$

Why? Double counts $Pr(A \& B)$

$$Pr(A \text{ OR } B) =$$

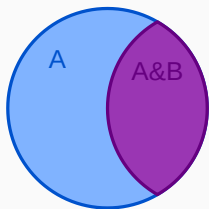
$$Pr(A) + Pr(B) - Pr(A \& B)$$

$$\Rightarrow Pr(A \text{ OR } B) =$$

$$0.2 + 0.3 - 0.1 = 0.4$$



Non-independent events



Calculating probabilities

$$Pr(B | A)$$

“The probability of observing B given that we already know A happened”

$$Pr(B | A) = \frac{Pr(A \& B)}{Pr(A)}$$

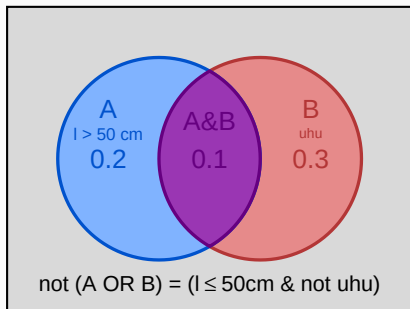
The **event** is $A \& B$ and all the possible **outcomes** are $A \& B$ or $A \text{ NOT } B$. So we need to divide the probability of $A \& B$ by the probability of $A \& B$ or $A \text{ NOT } B$.

And $Pr(A) = Pr(A \& B) + Pr(A \text{ NOT } B)$ so

$$Pr(B | A) = \frac{Pr(A \& B)}{Pr(A)}$$

Non-independent events

Calculating probabilities



Non-independent events

Calculating probabilities

Note that we can re-arrange this equation

$$Pr(B | A) = \frac{Pr(A \& B)}{Pr(A)}$$

$$Pr(B | A)Pr(A) = Pr(A \& B)$$

By extension, this equation is also true

$$Pr(A | B)Pr(B) = Pr(A \& B)$$

Therefore

$$Pr(A | B)Pr(B) = Pr(B | A)Pr(A)$$

Non-independent events

$$Pr(A | B)Pr(B) = Pr(B | A)Pr(A)$$

And re-arranging

$$Pr(A | B) = \frac{Pr(B | A)Pr(A)}{Pr(B)}$$

This is Bayes' Theorem and it is super useful

Rules of probability

1. Probability is always between 0 and 1
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3. We can *add* the probabilities of **mutually exclusive** events
4. We can *add* the probabilities of **independent** events
5. We can *simply multiply* the probabilities of **independent** events

For non-independent events (but also works for independent events):

6. $Pr(A \text{ OR } B) = Pr(A) + Pr(B) - Pr(A \& B)$
7. $Pr(A \& B) = Pr(A|B)Pr(B)$
8. Bayes' theorem: $Pr(A|B) = \frac{Pr(B|A)Pr(A)}{Pr(B)}$